

Package ‘lpmec’

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Title Measurement Error Analysis and Correction Under Identification Restrictions

Version 1.1.4

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Description Implements methods for analyzing latent variable models with measurement error correction, including Item Response Theory (IRT) models. Provides tools for various correction methods such as Bayesian Markov Chain Monte Carlo (MCMC), over-imputation, bootstrapping for robust standard errors, Ordinary Least Squares (OLS), and Instrumental Variables (IV) based approaches. Supports flexible specification of observable indicators and groupings for latent variable analyses in social sciences and other fields. Methods are described in a working paper (2025) <[doi:10.48550/arXiv.2507.22218](https://doi.org/10.48550/arXiv.2507.22218)>.

Depends R (>= 3.5.0)

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LazyData false

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stats,
sensemakr,
pscl,
AER,
sandwich,
mvtnorm,
Amelia,
emIRT,
gtools

Suggests testthat (>= 3.0.0),
knitr,
rmarkdown

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URL <https://github.com/cjerzak/lpmec-software>

BugReports <https://github.com/cjerzak/lpmec-software/issues>

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build_backend	<i>A function to build the environment for lpmec. Builds a conda environment in which 'JAX', 'numpyro', and 'numpy' are installed. Users can also create a conda environment where 'JAX' and 'numpy' are installed themselves.</i>
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Description

A function to build the environment for lpmec. Builds a conda environment in which 'JAX', 'numpyro', and 'numpy' are installed. Users can also create a conda environment where 'JAX' and 'numpy' are installed themselves.

Usage

```
build_backend(conda_env = "lpmec", conda = "auto")
```

Arguments

conda_env	(default = "lpmec") Name of the conda environment in which to place the backends.
conda	(default = auto) The path to a conda executable. Using "auto" allows reticulate to attempt to automatically find an appropriate conda binary.

Value

Invisibly returns NULL; this function is used for its side effects of creating and configuring a conda environment for lpmec. This function requires an Internet connection. You can find out a list of conda Python paths via: `Sys.which("python")`

Examples

```
## Not run:
# Create a conda environment named "lpmec"
# and install the required Python packages (jax, numpy, etc.)
build_backend(conda_env = "lpmec", conda = "auto")

# If you want to specify a particular conda path:
```

```
# build_backend(conda_env = "lpmec", conda = "/usr/local/bin/conda")

## End(Not run)
```

infer_orientation_signs

Infer orientation signs for each observable indicator

Description

This helper analyzes observable indicators and returns a numeric vector of 1 or -1 for use with the `orientation_signs` argument in `lpmec`. Each sign is chosen so that the correlation between the oriented indicator and either the outcome `Y` or the first principal component of the indicators is positive.

Usage

```
infer_orientation_signs(Y, observables, method = c("Y", "PC1"))
```

Arguments

<code>Y</code>	Numeric outcome vector. Only used when <code>method = "Y"</code> .
<code>observables</code>	A matrix or data frame of binary observable indicators.
<code>method</code>	Character string specifying how to orient the indicators. "Y" orient each indicator so that its correlation with <code>Y</code> is positive. "PC1" orient each indicator so that its correlation with the first principal component of <code>observables</code> is positive. Default is "Y".

Value

A numeric vector of length `ncol(observables)` containing 1 or -1.

Examples

```
set.seed(1)
Y <- rnorm(10)
obs <- data.frame(matrix(sample(c(0,1), 20, replace = TRUE), ncol = 2))
infer_orientation_signs(Y, obs)
```

KnowledgeVoteDuty	<i>KnowledgeVoteDuty: Survey Respondents' Views of Voting as a Duty and Political Knowledge Questions</i>
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Description

KnowledgeVoteDuty is a modified set of responses to a small set of questions on the American National Election Study's 2024 Time Series Study. These data only include respondents who had non-missing values on all of the variables included, dropping respondents with one or more missing values.

Usage

```
data(KnowledgeVoteDuty)
```

Format

A data frame with 3,059 observations and 5 variables:

voteduty Whether respondents feel that voting is a duty or a choice. Values range from 1 to 7, with 1 being "Very strongly a duty" and 7 being "Very strongly a choice," created based on variable V241218x.

SenateTerm Dummy variable (0 or 1) for whether respondent correctly stated the length of a U.S. Senate term. Created based on variable V241612.

SpendLeast Dummy variable (0 or 1) for whether respondent correctly identified "Foreign aid" from a list as the category the federal government spends the least on. Created based on variable V241613.

HouseParty Dummy variable (0 or 1) for whether respondent correctly identified the political party that currently has the most members in the U.S. House of Representatives. Created based on variable V241614.

SenateParty Dummy variable (0 or 1) for whether respondent correctly identified the political party that currently has the most members in the U.S. Senate. Created based on variable V241615.

References

American National Election Studies. 2024. ANES 2024 Time Series Study Full Release [dataset and documentation]. Available at electionstudies.org.

Examples

```
data(KnowledgeVoteDuty)
voteduty <- KnowledgeVoteDuty$voteduty
knowledge <- scale(rowMeans(KnowledgeVoteDuty[, -1]))
summary(lm(voteduty ~ knowledge))
```

lpmec

lpmec

Description

Implements latent variable models with measurement error correction

Usage

```
lpmec(
  Y,
  observables,
  observables_groupings = colnames(observables),
  orientation_signs = NULL,
  make_observables_groupings = FALSE,
  n_boot = 32L,
  n_partition = 10L,
  partition_aggregation = "median",
  partition_aggregation_probs = c(0.01, 0.99),
  boot_basis = 1:length(Y),
  return_intermediaries = TRUE,
  ordinal = FALSE,
  estimation_method = "em",
  latent_estimation_fn = NULL,
  mcmc_control = list(backend = "pscl", n_samples_warmup = 500L, n_samples_mcmc = 1000L,
    batch_size = 512L, chain_method = "parallel", subsample_method = "full",
    anchor_parameter_id = NULL, n_thin_by = 1L, n_chains = 2L, outcome_prior =
    list(calibration = "data"), joint2_prior = list()),
  conda_env = "lpmec",
  conda_env_required = FALSE
)
```

Arguments

<code>Y</code>	A vector of observed outcome variables
<code>observables</code>	A matrix of observable indicators used to estimate the latent variable
<code>observables_groupings</code>	A vector specifying groupings for the observable indicators. Default is column names of observables.
<code>orientation_signs</code>	(optional) A numeric vector of length equal to the number of columns in ‘observables’, containing 1 or -1 to indicate the desired orientation of each column. If provided, each column of ‘observables’ will be oriented by this sign before analysis. Default is NULL (no orientation applied).
<code>make_observables_groupings</code>	Logical. If TRUE, creates dummy variables for each level of the observable indicators. Default is FALSE.
<code>n_boot</code>	Non-negative integer. Number of bootstrap iterations. Use 0 to disable bootstrap resampling and fit only the original sample. Default is 32.

<code>n_partition</code>	Positive integer. Number of split-half partitions for each bootstrap iteration. When <code>n_boot = 0</code> , this still controls how many original-sample partition runs are aggregated. Default is 10.
<code>partition_aggregation</code>	Aggregation strategy for combining estimates across partitions within each bootstrap iteration. Default is "median". Options are "median", "winsorized_mean", "trimmed_mean", or a custom function that accepts a numeric vector and returns one numeric value.
<code>partition_aggregation_probs</code>	Numeric vector of length 2 used by "winsorized_mean" and "trimmed_mean". For winsorization, values are clipped to these quantiles before averaging. For trimming, values outside these quantiles are dropped before averaging. Default is <code>c(0.01, 0.99)</code> .
<code>boot_basis</code>	Vector of indices or grouping variable for stratified bootstrap. Default is <code>1:length(Y)</code> .
<code>return_intermediaries</code>	Logical. If TRUE, returns intermediate results. Default is TRUE.
<code>ordinal</code>	Logical indicating whether the observable indicators are ordinal (TRUE) or binary (FALSE).
<code>estimation_method</code>	Character specifying the estimation approach. Options include: "em" (default): Uses expectation-maximization via <code>emIRT</code> package. Supports both binary (via <code>emIRT::binIRT</code>) and ordinal (via <code>emIRT::ordIRT</code>) indicators. "pca": First principal component of observables. "averaging": Uses feature averaging. "mcmc": Markov Chain Monte Carlo estimation using either <code>pscl::ideal</code> (R backend) or <code>numpyro</code> (Python backend) "mcmc_joint": Joint Bayesian model that simultaneously estimates latent variables and outcome relationship using <code>numpyro</code> "mcmc_joint2": NumPyro mixed factor-analysis benchmark with binary indicators and continuous Y in one factor model "mcmc_overimputation": Two-stage MCMC approach with measurement error correction via over-imputation "custom": In this case, latent estimation performed using <code>latent_estimation_fn</code>.
<code>latent_estimation_fn</code>	Custom function for estimating latent trait from observables if <code>estimation_method="custom"</code> (optional). The function should accept a matrix of observables (rows are observations) and return a numeric vector of length equal to the number of observations.
<code>mcmc_control</code>	A list indicating parameter specifications if MCMC used. <code>backend</code> Character string indicating the MCMC engine to use. Valid options are "pscl" (default, uses the R-based <code>pscl::ideal</code> function) or "numpyro" (uses the Python <code>numpyro</code> package via <code>reticulate</code>). <code>n_samples_warmup</code> Integer specifying the number of warm-up (burn-in) iterations before samples are collected. Default is 500. <code>n_samples_mcmc</code> Integer specifying the number of post-warmup MCMC iterations to retain. Default is 1000. <code>batch_size</code> Integer row subsample size used when <code>subsample_method="batch"</code> with the NumPyro backend. Must be between 1 and <code>nrow(observables) - 1</code>. Default is 512.

	<code>subsample_method</code> Character string for NumPyro likelihood evaluation. Use "full" (default) for all rows or "batch" for experimental HMCECS row subsampling with <code>batch_size</code>. <code>chain_method</code> Character string passed to numpyro specifying how to run multiple chains. Options: "parallel" (default), "sequential", or "vectorized". <code>anchor_parameter_id</code> Optional 1-based observable-column index used by NumPyro MCMC backends to anchor item difficulty orientation. If omitted or NULL, automatic orientation is used. <code>n_thin_by</code> Integer indicating the thinning factor for MCMC samples. Default is 1. <code>n_chains</code> Integer specifying the number of parallel MCMC chains to run. Default is 2. <code>outcome_prior</code> List controlling "mcmc_joint" outcome-model priors for the NumPyro backend. By default, <code>calibration = "data"</code> centers the intercept prior at $\text{mean}(Y)$ and scales intercept, slope, and residual-sigma priors by $\text{sd}(Y)$. Use <code>calibration = "legacy"</code> to restore the previous unit-scale priors, or provide numeric overrides for <code>intercept_mean</code>, <code>intercept_sd</code>, <code>slope_mean</code>, <code>slope_sd</code>, and <code>sigma_sd</code>. The optional <code>scale_floor</code> sets the minimum scale used for data-calibrated priors. <code>joint2_prior</code> List controlling "mcmc_joint2" priors. Defaults are <code>lambda_mean = 0</code>, <code>lambda_sd = 2</code>, <code>psi_shape = 0.0005</code>, and <code>psi_scale = 0.0005</code>. For item loadings, <code>lambda_mean</code> and <code>lambda_sd</code> parameterize the raw loading scale before the positive softplus transform; all item loadings are positive by default.
<code>conda_env</code>	A character string specifying the name of the conda environment to use via reticulate. Default is "lpmec".
<code>conda_env_required</code>	A logical indicating whether the specified conda environment must be strictly used. If TRUE, an error is thrown if the environment is not found. Default is FALSE.

Details

This function implements a latent variable analysis with measurement error correction. It fits the original sample and, when `n_boot >= 1`, performs bootstrap resampling for uncertainty estimates. Each original or bootstrap sample is analyzed with one or more split-half partitions. For each partition, it calls the `lpmec_onerun` function to estimate latent variables and apply various correction methods. The results are then aggregated across partitions and bootstrap iterations to produce final estimates and, when bootstrap draws are available, bootstrap standard errors.

Value

A list containing various estimates and statistics (in `snake_case`):

- Naive, IV, corrected IV, and corrected OLS estimates: `ols_*`, `iv_*`, `corrected_iv_*`, and `corrected_ols_*`. Bootstrap uncertainty summaries use suffixes `_se`, `_lower`, `_upper`, and `_tstat` where applicable.
- `var_est_split` and `var_est_split_se`: Aggregated split-half measurement-error variance and, when bootstrap draws are available, its bootstrap standard error.
- `bayesian_ols_*_outer_normed` and `bayesian_ols_*_inner_normed`: MCMC coefficient summaries. The `*_parametric` standard-error fields retain within-run posterior uncertainty, while the non-parametric standard-error and interval fields summarize bootstrap variation.

- `m_stage_1_erv*` and `m_reduced_erv*`: Extreme robustness values and bootstrap uncertainty summaries for the first-stage and reduced-form regressions.
- `mcmc_joint2*`: NumPyro "mcmc_joint2" diagnostics, including effective-sample-size percentages, maximum R-hat, divergent transitions, mean accept probability, and orientation diagnostics.
- `x_est1` and `x_est2`: Split-half latent variable estimates from the original sample.
- `Intermediary_*`: Per-run original-sample and bootstrap outputs, returned only when `return_intermediaries = TRUE`.

References

Jerzak, C. T. and Jessee, S. A. (2025). Attenuation Bias with Latent Predictors. arXiv:2507.22218 [stat.AP]. <https://arxiv.org/abs/2507.22218>

Examples

```
# Generate some example data
set.seed(123)
Y <- rnorm(1000)
observables <- as.data.frame(matrix(sample(c(0,1), 1000*10, replace = TRUE), ncol = 10))

# Run the bootstrapped analysis
results <- lpmec(Y = Y,
  observables = observables,
  n_boot = 10,    # small values for illustration only
  n_partition = 5 # small for size
)

# Use a winsorized mean across partitions
results_winsorized <- lpmec(Y = Y,
  observables = observables,
  n_boot = 10,
  n_partition = 5,
  partition_aggregation = "winsorized_mean")

# View the corrected IV coefficient and its standard error
print(results)
```

lpmec_onerun

lpmec_onerun

Description

Implements analysis for latent variable models with measurement error correction

Usage

```
lpmec_onerun(
  Y,
  observables,
```



```

observables_groupings = colnames(observables),
make_observables_groupings = FALSE,
estimation_method = "em",
latent_estimation_fn = NULL,
mcmc_control = list(backend = "pscl", n_samples_warmup = 500L, n_samples_mcmc = 1000L,
  batch_size = 512L, chain_method = "parallel", subsample_method = "full", n_thin_by =
  1L, n_chains = 2L, outcome_prior = list(calibration = "data"), joint2_prior = list()),
ordinal = FALSE,
conda_env = "lpmec",
conda_env_required = FALSE
)

```

Arguments

- Y** A vector of observed outcome variables
- observables** A matrix of observable indicators used to estimate the latent variable
- observables_groupings** A vector specifying groupings for the observable indicators. Default is column names of observables.
- make_observables_groupings** Logical. If TRUE, creates dummy variables for each level of the observable indicators. Default is FALSE.
- estimation_method** Character specifying the estimation approach. Options include:
- "em" (default): Uses expectation-maximization via emIRT package. Supports both binary (via `emIRT::binIRT`) and ordinal (via `emIRT::ordIRT`) indicators.
 - "pca": First principal component of observables.
 - "averaging": Uses feature averaging.
 - "mcmc": Markov Chain Monte Carlo estimation using either `pscl::ideal` (R backend) or `numpyro` (Python backend)
 - "mcmc_joint": Joint Bayesian model that simultaneously estimates latent variables and outcome relationship using `numpyro`
 - "mcmc_joint2": NumPyro mixed factor-analysis benchmark with binary indicators and continuous Y in one factor model
 - "mcmc_overimputation": Two-stage MCMC approach with measurement error correction via over-imputation
 - "custom": In this case, latent estimation performed using `latent_estimation_fn`.
- latent_estimation_fn** Custom function for estimating latent trait from observables if `estimation_method="custom"` (optional). The function should accept a matrix of observables (rows are observations) and return a numeric vector of length equal to the number of observations.
- mcmc_control** A list indicating parameter specifications if MCMC used.
- backend** Character string indicating the MCMC engine to use. Valid options are "pscl" (default, uses the R-based `pscl::ideal` function) or "numpyro" (uses the Python `numpyro` package via `reticulate`).
- n_samples_warmup** Integer specifying the number of warm-up (burn-in) iterations before samples are collected. Default is 500.

	<code>n_samples_mcmc</code> Integer specifying the number of post-warmup MCMC iterations to retain. Default is 1000.
	<code>batch_size</code> Integer row subsample size used when <code>subsample_method = "batch"</code> with the NumPyro backend. Must be between 1 and <code>nrow(observables) - 1</code> . Default is 512.
	<code>subsample_method</code> Character string for NumPyro likelihood evaluation. Use "full" (default) for all rows or "batch" for experimental HMCECS row subsampling with <code>batch_size</code> .
	<code>chain_method</code> Character string passed to numpyro specifying how to run multiple chains. Options: "parallel" (default), "sequential", or "vectorized".
	<code>anchor_parameter_id</code> Optional 1-based observable-column index used by NumPyro MCMC backends to anchor item difficulty orientation. If omitted or NULL, automatic orientation is used.
	<code>n_thin_by</code> Integer indicating the thinning factor for MCMC samples. Default is 1.
	<code>n_chains</code> Integer specifying the number of parallel MCMC chains to run. Default is 2.
	<code>outcome_prior</code> List controlling "mcmc_joint" outcome-model priors for the NumPyro backend. By default, <code>calibration = "data"</code> centers the intercept prior at <code>mean(Y)</code> and scales intercept, slope, and residual-sigma priors by <code>sd(Y)</code> . Use <code>calibration = "legacy"</code> to restore the previous unit-scale priors, or provide numeric overrides for <code>intercept_mean</code> , <code>intercept_sd</code> , <code>slope_mean</code> , <code>slope_sd</code> , and <code>sigma_sd</code> . The optional <code>scale_floor</code> sets the minimum scale used for data-calibrated priors.
	<code>joint2_prior</code> List controlling "mcmc_joint2" priors. Defaults are <code>lambda_mean = 0</code> , <code>lambda_sd = 2</code> , <code>psi_shape = 0.0005</code> , and <code>psi_scale = 0.0005</code> . For item loadings, <code>lambda_mean</code> and <code>lambda_sd</code> parameterize the raw loading scale before the positive softplus transform; all item loadings are positive by default.
<code>ordinal</code>	Logical indicating whether the observable indicators are ordinal (TRUE) or binary (FALSE).
<code>conda_env</code>	A character string specifying the name of the conda environment to use via reticulate. Default is "lpmec".
<code>conda_env_required</code>	A logical indicating whether the specified conda environment must be strictly used. If TRUE, an error is thrown if the environment is not found. Default is FALSE.

Details

This function implements a latent variable analysis with measurement error correction. It splits the observable indicators into two sets, estimates latent variables using each set, and then applies various correction methods including OLS correction and instrumental variable approaches.

Value

A list containing various estimates and statistics:

- Naive, IV, corrected IV, and corrected OLS estimates: `ols_*`, `iv_*`, `corrected_iv_*`, and `corrected_ols_*`. Split-specific estimates use suffixes `_a` and `_b`.
- `var_est_split`: Estimated split-half measurement-error variance.

- `bayesian_ols*_outer_normed` and `bayesian_ols*_inner_normed`: MCMC coefficient summaries when an MCMC method is used; otherwise NA.
- `m_stage1_erv` and `m_reduced_erv`: Extreme robustness values for the first-stage and reduced-form regressions.
- `outcome_prior_*`: Resolved outcome-prior values used by NumPyro joint outcome models.
- `mcmc_joint2_*`: NumPyro "mcmc_joint2" diagnostics, including effective-sample-size percentages, maximum R-hat, divergent transitions, mean accept probability, and orientation diagnostics.
- `x_est1` and `x_est2`: Split-half latent variable estimates.

Standard Errors

The following single-run standard errors and t-statistics are currently returned as NA because their analytical derivation is not yet implemented:

- `corrected_iv_se`: Standard error for the corrected IV coefficient
- `corrected_ols_se`: Standard error for the corrected OLS coefficient
- `corrected_ols_tstat`: T-statistic for the corrected OLS coefficient
- `corrected_ols_coef_alt`: Alternative corrected OLS coefficient

For inference on these quantities, use the bootstrap approach via [lpmec](#), which provides valid confidence intervals and standard errors through resampling.

Examples

```
# Generate some example data
set.seed(123)
Y <- rnorm(1000)
observables <- as.data.frame(matrix(sample(c(0,1), 1000*10, replace = TRUE), ncol = 10))

# Run the analysis
results <- lpmec_onerun(Y = Y,
                        observables = observables)

# View the corrected estimates
print(results)
```

plot.lpmec

Plot method for lpmec objects

Description

Creates visualizations of LPMEC model results. Can plot either the latent variable estimates or the bootstrap distribution of coefficients.

Usage

```
## S3 method for class 'lpmec'
plot(x, type = "latent", ...)
```

Arguments

x	An object of class lpmec returned by lpmec .
type	Character string specifying the plot type. Either "latent" (default) for a scatter plot of split-half latent estimates, or "coefficients" for a density plot of bootstrap coefficient estimates.
...	Additional arguments passed to plot or density .

Value

No return value, called for side effects (creates a plot).

See Also

[lpmec](#), [summary.lpmec](#), [print.lpmec](#)

plot.lpmec_onerun	<i>Plot method for lpmec_onerun objects</i>
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Description

Creates a scatter plot comparing the two split-half latent variable estimates.

Usage

```
## S3 method for class 'lpmec_onerun'
plot(x, ...)
```

Arguments

x	An object of class lpmec_onerun returned by lpmec_onerun .
...	Additional arguments passed to plot .

Value

No return value, called for side effects (creates a plot).

See Also

[lpmec_onerun](#), [summary.lpmec_onerun](#), [print.lpmec_onerun](#)

print.lpmec	<i>Print method for lpmec objects</i>
-------------	---------------------------------------

Description

Prints a concise summary of bootstrapped LPMEC model results.

Usage

```
## S3 method for class 'lpmec'  
print(x, ...)
```

Arguments

x	An object of class lpmec returned by lpmec .
...	Additional arguments (currently unused).

Value

The input object x, returned invisibly.

See Also

[lpmec](#), [summary.lpmec](#), [plot.lpmec](#)

print.lpmec_onerun	<i>Print method for lpmec_onerun objects</i>
--------------------	--

Description

Prints a concise summary of single-run LPMEC model results.

Usage

```
## S3 method for class 'lpmec_onerun'  
print(x, ...)
```

Arguments

x	An object of class lpmec_onerun returned by lpmec_onerun .
...	Additional arguments (currently unused).

Value

The input object x, returned invisibly.

See Also

[lpmec_onerun](#), [summary.lpmec_onerun](#), [plot.lpmec_onerun](#)

summary.lpmec	<i>Summary method for lpmec objects</i>
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Description

Provides a comprehensive summary of bootstrapped LPMEC model results including OLS, IV, corrected, and Bayesian coefficient estimates with confidence intervals.

Usage

```
## S3 method for class 'lpmec'
summary(object, ...)
```

Arguments

object	An object of class lpmec returned by lpmec .
...	Additional arguments (currently unused).

Value

A data frame containing coefficient estimates, standard errors, and confidence intervals, returned invisibly. The data frame has rows for OLS, IV, Corrected IV, Corrected OLS, and Bayesian OLS estimates.

See Also

[lpmec](#), [print.lpmec](#), [plot.lpmec](#)

summary.lpmec_onerun	<i>Summary method for lpmec_onerun objects</i>
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Description

Provides a summary of single-run LPMEC model results including OLS, IV, and corrected coefficient estimates.

Usage

```
## S3 method for class 'lpmec_onerun'
summary(object, ...)
```

Arguments

object	An object of class lpmec_onerun returned by lpmec_onerun .
...	Additional arguments (currently unused).

Value

A data frame containing coefficient estimates and standard errors, returned invisibly. The data frame has rows for OLS, IV, Corrected IV, and Corrected OLS estimates.

See Also

[lpmec_onerun](#), [print.lpmec_onerun](#), [plot.lpmec_onerun](#)

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